

LUKAS FENKAM

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OBJECTIVE

Computer Science student seeking a co-op in data science, software development or AI research, shaped by hands-on experience designing, evaluating, and deploying systems. Motivated by applying research-driven thinking and practical engineering skills to build reliable, responsible, and trustworthy AI technologies.

HIGHLIGHTS OF SKILLS & QUALIFICATIONS

- Python machine learning and software development, training CNN image classifiers achieving 80%+ accuracy.
- Linux systems, Java software development, Python automation, Ansible, Git, GitHub, Flutter, Android Studio.
- HTML, CSS, and JavaScript web development, deploying sites via GitHub Pages and custom domains.
- Collaboration and documentation using Microsoft Word, Excel, PowerPoint, and Office 365.
- Trilingual: English, French and German

EDUCATION

Ontario Tech University

Oshawa, ON

Bachelor of Science Honours, Computer Science

Sept 2023 – Jun 2027 Expected

- Relevant courses: Data Structures, Algorithms, Machine Learning, Computer Vision, and Computer Graphics.
- Certifications: LPI Linux Essentials; IBM Generative AI Prompt Engineering and Applications.

PROJECT AND RESEARCH EXPERIENCE

OpenUSD / NVIDIA Omniverse Learning Project

2026

github.com/Luke1432/omniverse-openusd-learning

- Authored OpenUSD scenes programmatically using Python, creating stages, prim hierarchies, and geometric assets.
- Implemented USD composition using references and overrides to separate base assets from composed scenes.
- Applied transforms via USD schemas and validated results using `usdview`.
- Debugged OpenGL and GPU rendering issues on Linux, verifying hardware acceleration on NVIDIA GPUs.
- Documented reproducible development environments and workflows using micromamba.

Research Intern

Jul 2025 – Aug 2025

Trustworthy AI Lab, Ontario Tech University

Oshawa, ON

- Trained CNN models for forensic bloodstain pattern classification, achieving over 80% accuracy.
- Preprocessed and augmented 100+ images in Python to improve generalization and reduce overfitting.
- Collaborated with graduate researchers on iterative experiments and shared evaluation benchmarks.

Portfolio Website

2025 – Present

Independent Project

- Built and maintained a personal portfolio using HTML, CSS, and JavaScript to showcase projects and experience.
- Deployed website via GitHub Pages with a custom domain and responsive layout for mobile and desktop.
- Implemented responsive, platform-aware design to ensure consistent behavior across desktop and mobile devices.

Distributed Raspberry Pi Image Server Cluster

Dec 2025

Independent Project

- Built a Linux-based Raspberry Pi cluster using Python and Ansible to automate configuration across four nodes.
- Implemented Samba, NFS, rsync, and lsyncd for fault-tolerant synchronization of thousands of images.
- Automated ingestion and conversion pipelines in Python, reducing manual processing by over 90%.

Software Engineering Intern

Jul 2021

Red Hat Inc.

Remote

- Developed a Java-based GitHub automation bot using Quarkus to streamline issue and pull request workflows.
- Reduced manual triage effort by an estimated 20–30% through automated labeling and workflow rules.

VOLUNTEERING

Organizer and Workshop Leader

Jan 2024 – Present

Aproresco HackApro Hackathon

Oshawa, ON

- Organized an annual bilingual hackathon for over 50 participants, managing logistics, evaluation, and team support.
- Led Scratch programming workshops for middle and high school students to increase engagement.